

NPLify · P0 Technical Baseline

Deliverable 1 — Entity-Relationship Diagram & Data Model · Draft v4.0

New XP Technologies Limited

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Status: Rebuilt from ERD v3.0 (2 October 2026), the *NPLify — Project Understanding v1.0* with Sud's review comments, and the ERD review of 6–7 October 2026 that traced three operational scenarios (a payout rejected by the receiving bank, a two-leg route across two partners, an ad-hoc conversion of a client balance) against the model. Supersedes v3.0 in full.

Convention: all monetary amounts stored as exact decimals; every table carries `created_at` / `created_by`; audit, approvals and locking are modelled once (domain 4) and apply to all master data. Derived figures (client balance, group entitlement, offsets, custody) are **views over postings and deals** — **never stored**.

Six diagrams across four domains, one schema: 55 tables and 4 views. Grey dashed boxes in a diagram mark entities defined in another diagram, and name it. Arrows run in the direction of data flow, from the referenced row into the row that stores the foreign key; a filled arrow is composition (the parent owns the child), an open arrow a reference, a dashed line a polymorphic link resolved by a type column plus an id. Section 6 lists what changed since v3.0; Appendix A reconciles the model with the FRS; Appendix B is the data dictionary.

1 · Configuration domain

A **Project** (= client) owns its configuration. Partners are global counterparties; how a partner is used *in this project* is the **partner configuration** block — fees, vehicles, endpoints, registrations and rebate terms hang off it. Receivers are organised in **receiver groups**, the unit entitlement is tracked on, and each receiving entity holds one or more **accounts** with a currency and rail.

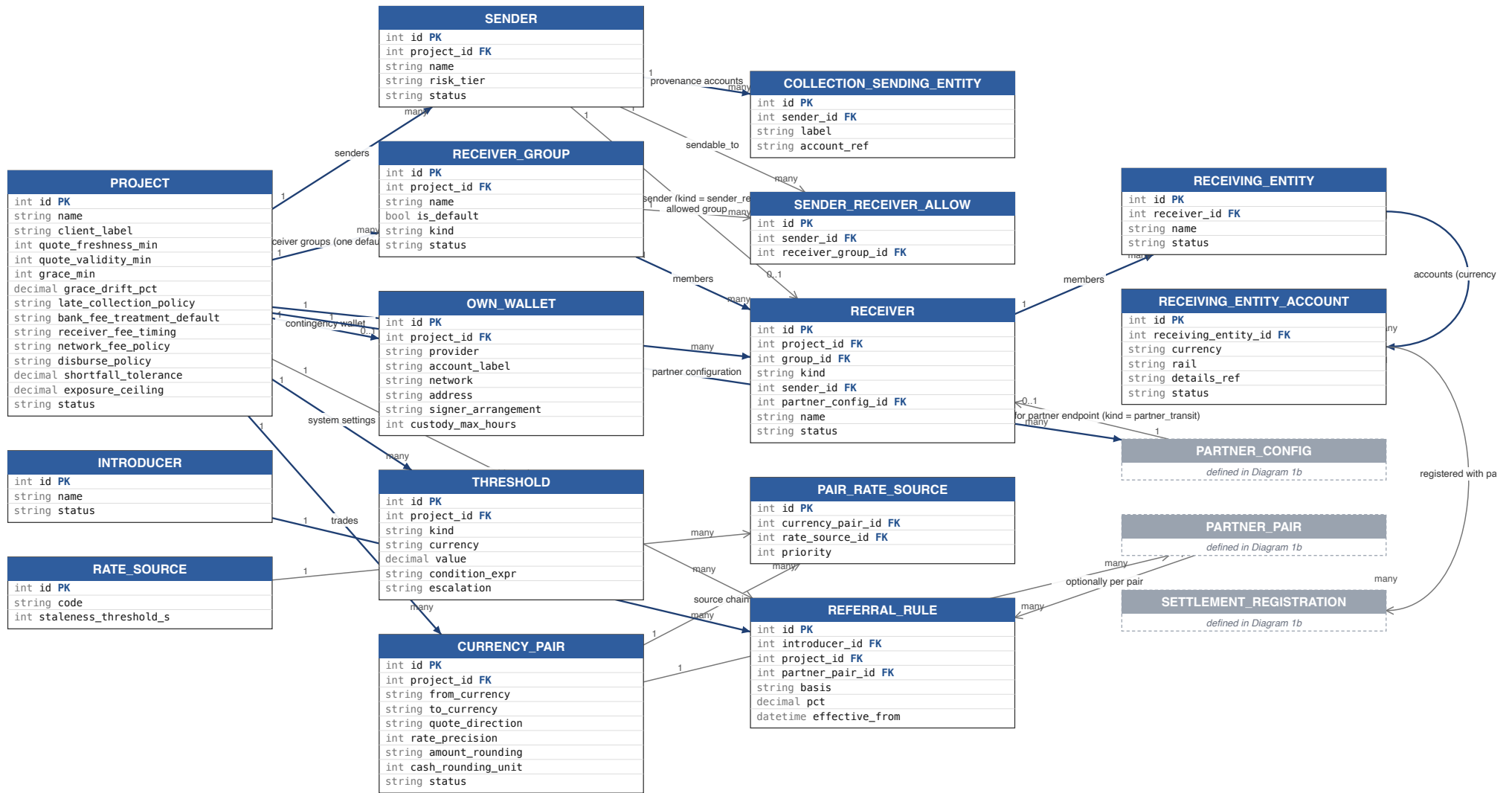


Diagram 1a — Configuration — project, parties and receiver groups

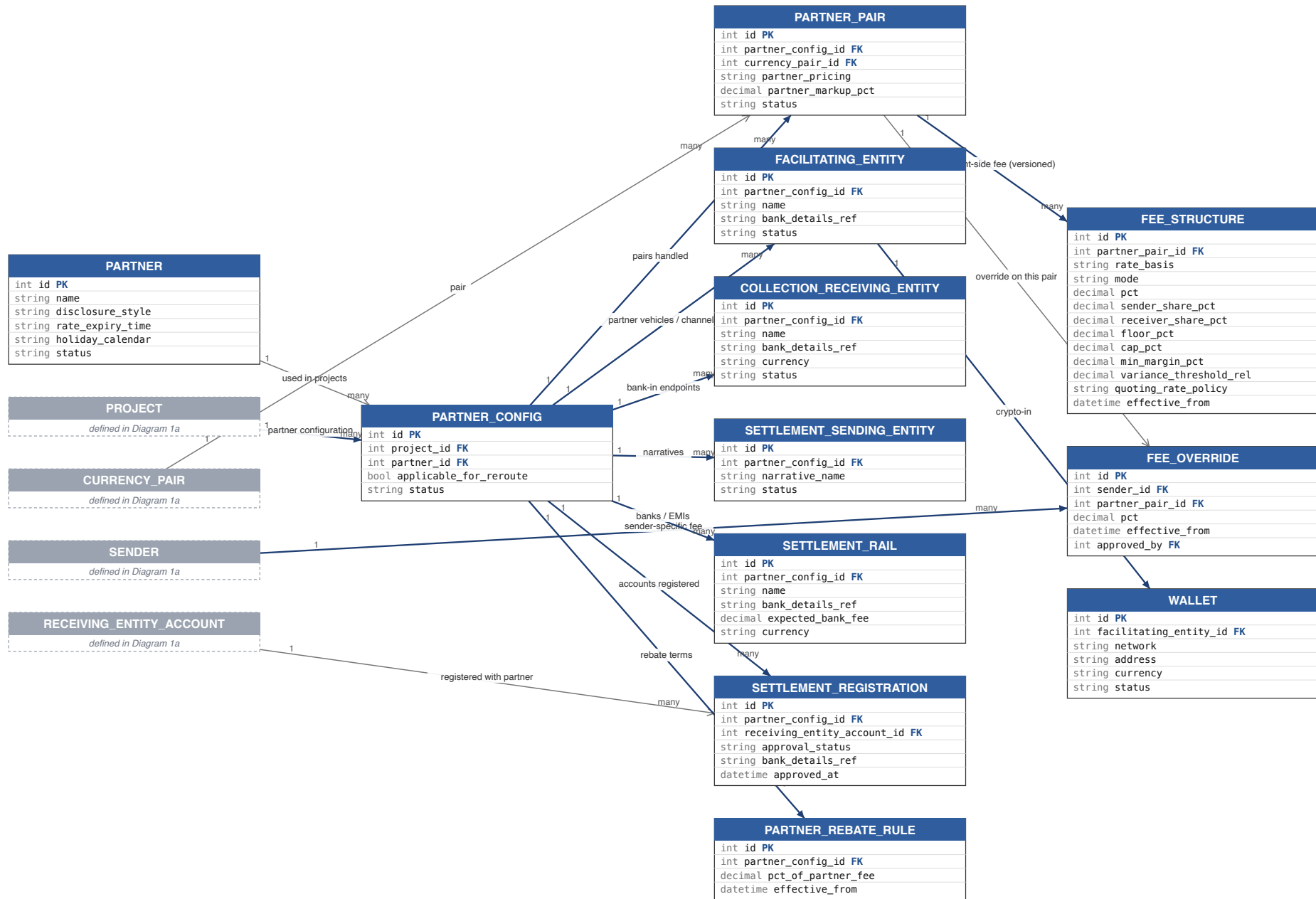


Diagram 1b — Configuration — partner configuration block

Notes

- A **Project** (= client) owns its configuration. Receivers sit in **receiver groups**: every project has one default group and every receiver belongs to exactly one group. Entitlement is tracked per group, so payouts within a group are ordinary settlement and only payouts across groups are offsets; a project with a single group never has an offset.
- A **Receiver** has a kind: counterparty for a client receiver; sender_return stands for a sender so refunds use the ordinary payout path and the partner's destination approval; partner_transit stands for a partner's collection endpoint so the hop of a two-leg route is an ordinary payout. sender_return receivers live in the project's return group and never appear on forward deals or allow lists.
- A receiving entity holds one or more **accounts**, each with a currency and rail (EUR via SWIFT and EUR via SEPA are two accounts). Registrations and payout lines point at the account.
- Fees are per partner × pair (FEE_STRUCTURE under PARTNER_PAIR), so choosing the partner is part of pricing. PARTNER.disclosure_style records whether the partner's markup is visible; partner_markup_pct is known only for market_plus partners.
- Three collection-endpoint kinds by method: crypto-in → WALLET; bank-in → COLLECTION_RECEIVING_ENTITY; cash-in → no record (instructions on the deal). Cash collections round to the nearest 100 / 500 / 1000 units by currency (CURRENCY_PAIR.cash_rounding_unit).
- Destination approval is per partner: SETTLEMENT_REGISTRATION is an account of the project's receiving entity as registered with this partner. SETTLEMENT_SENDING_ENTITY is the narrative on an outgoing transfer; SETTLEMENT_RAIL is the bank/EMI that executes it; neither holds money. COLLECTION_SENDING_ENTITY is provenance only and never a payout destination.
- OWN_WALLET exists for rerouting and pass-through (AQN-LT-Sub is an own wallet at Aquanow); PARTNER_CONFIG.applicable_for_reroute defines the alternate-partner set. INTRODUCER / REFERRAL_RULE record referral commissions NPL pays; PARTNER_REBATE_RULE records rebates a partner pays NPL.

2 · Rates domain

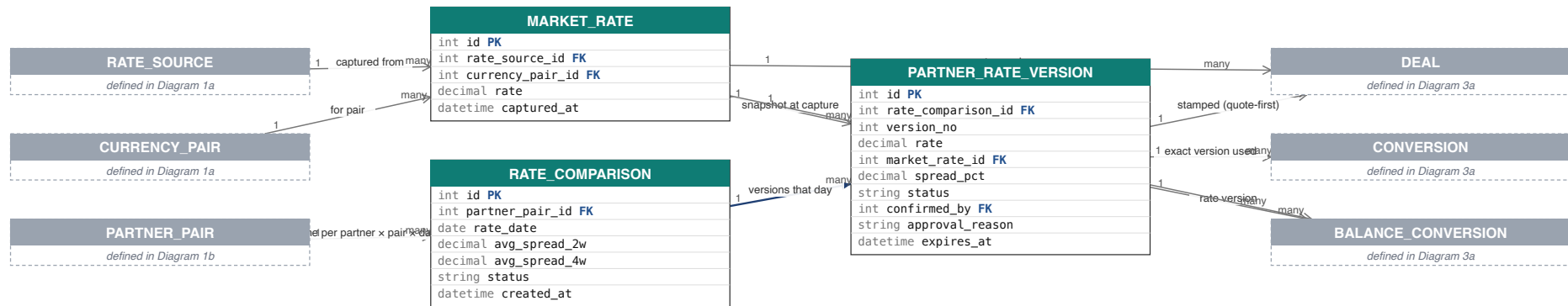


Diagram 2 — Rates

Notes

- The rate comparison for the day is created lazily by the first deal that needs it, one per partner × pair × date, and reused by every deal that day. A partner may give a new rate during the day, so one date can hold several versions. Rolling averages are built from confirmed versions only.
- A version is breached when $\text{spread} > \text{avg_2w} \times (1 + \text{variance_threshold_rel})$; quoting and conversion on that partner × pair freeze until Finance approves with a reason, rejects, or asks for a better rate. Versions are never overwritten; every deal leg, conversion and balance conversion references its exact version.
- Cross-source slippage is a query over MARKET_RATE across sources for the same pair and window.

3 · Deal lifecycle domain

3a · Deal group, legs, collection, conversion, reroute, balance conversion

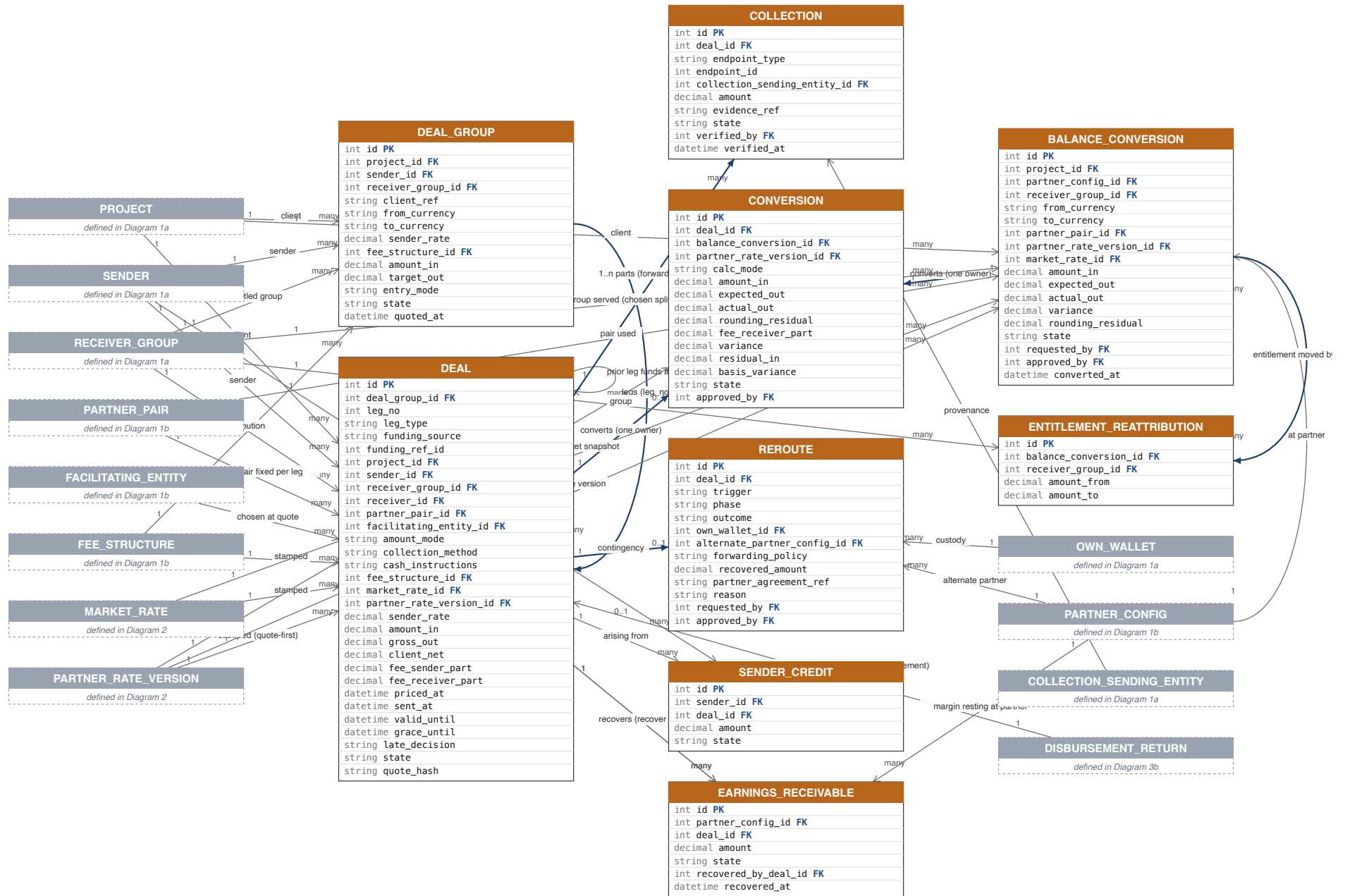


Diagram 3a – Deal lifecycle – deal group, legs, collection, conversion, reroute, balance conversion

3b · Disbursement, return, confirmation, fees, referrals and rebates

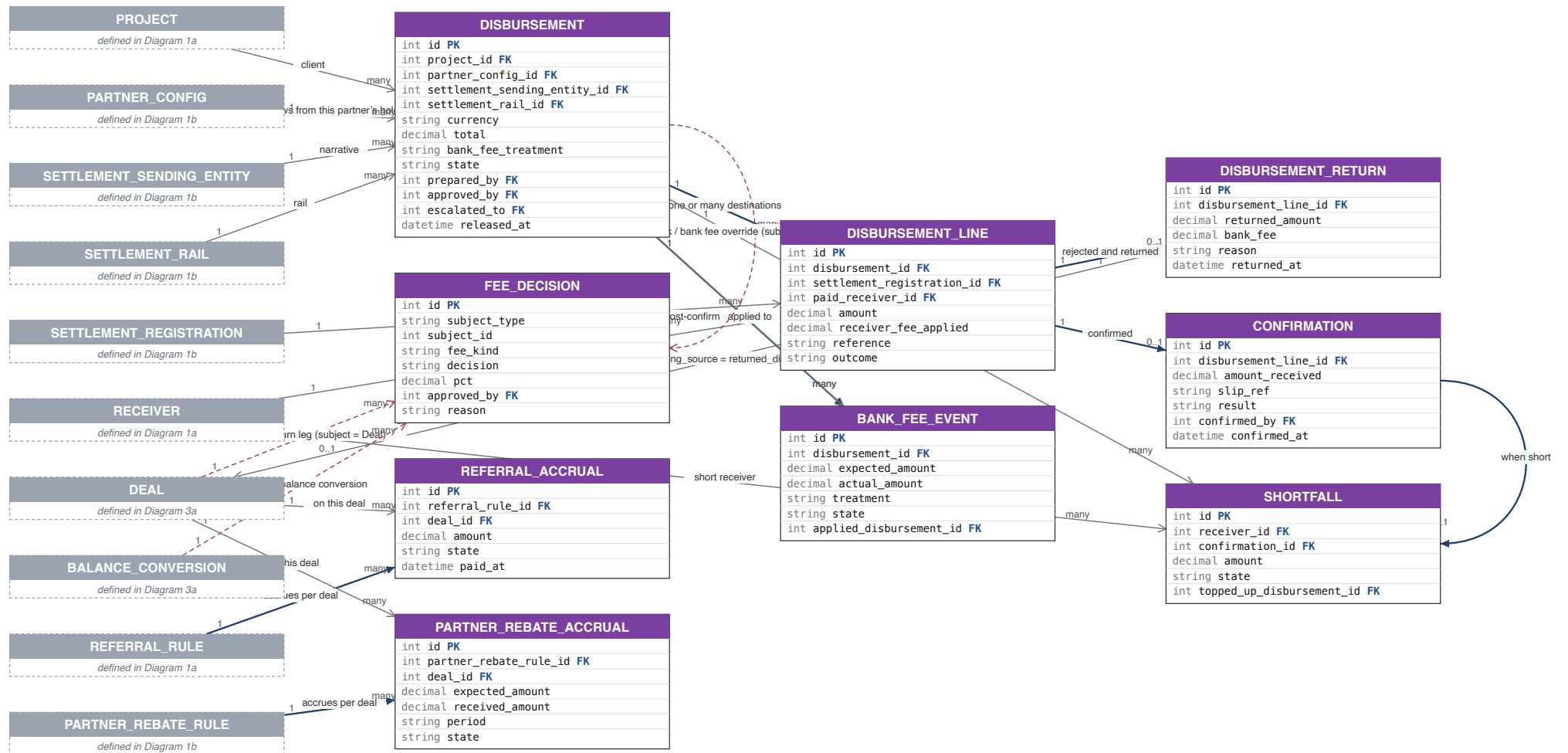


Diagram 3b — Disbursement, return, confirmation, fees, referrals and rebates

Derived views (never stored)

View	Definition
CLIENT_BALANCE(project, partner, currency)	CLIENT_BALANCE(project, partner, currency) = Σ conversions' actual_out booked at that partner (deal legs and balance conversions, target side) – Σ balance conversions' amount_in (source side) + Σ returned amounts – Σ released disbursement totals. Carries rounding dust below one unit.
GROUP_ENTITLEMENT(group, currency)	GROUP_ENTITLEMENT(group, currency) = Σ client_net of converted forward legs attributed to the group $\pm$ entitlement re-attributions – Σ DISBURSEMENT_LINE.amount paid to receivers in the group + Σ returned amounts. Replaces RECEIVER_ENTITLEMENT. Σ over a project's groups = Σ CLIENT_BALANCE over partners, per currency.
OFFSET(group, currency)	OFFSET(group, currency) = GROUP_ENTITLEMENT; positive = owed, negative = holding another group's flow. Can only be non-zero in a project with more than one group. Carries Management approval and tighter aging.
CUSTODY(own_wallet)	CUSTODY(own_wallet) = Σ collections into the own wallet under open reroutes + Σ recovered dues resting there – Σ forwarded; each row with deal, reason, age. The standing control on contingency holding.

Notes

- **Every deal belongs to a deal group.** DEAL_GROUP is the client-facing instruction: the pair as the sender sees it, the sender rate, fee, reference and entry mode. A plain single-partner deal is a group of one leg; a route that needs two partners (INR $\rightarrow$ USDT at Ali, USDT $\rightarrow$ EUR at Jeton) has two forward legs; a rejected payout is unwound with a return leg.
- **Legs.** leg_type forward or return; funding_source collection, prior_leg (leg 2 of a route, no collection) or returned_disbursement (a return leg). Priced once and partner fixed at quote hold per leg. Attribution is to the receiver group; the receiver is optional until settlement.
- Two entry modes as before: collect-first legs are born in state Collected with priced_at = conversion time; quote-first legs carry sent / valid / grace timestamps and a late decision. An estimate-first mode gives an indicative figure and prices at conversion.
- **Conversion has exactly one owner:** a deal leg or a BALANCE_CONVERSION. The receiver-side fee is taken at conversion. Amounts are truncated to whole units at conversion; the dropped fraction (rounding_residual) stays client money in the balance and pays out when it reaches a whole unit. NPL's margin is captured at conversion and is never again at risk from a return or a forward.
- **Balance conversion** is NPL converting a client's held balance (5K SGD $\rightarrow$ USDT to fund a 15K USDT payout): Finance-initiated, no sender, receiver or collection, same rate machinery, markup by FEE_DECISION. It names the receiver group it serves; ENTITLEMENT_REATTRIBUTION moves that group's entitlement to the new currency, and any excess over the group's entitlement is booked as an approved offset.
- **Reroute** is the only way partner or facilitating entity change after quote. Forwarding is forward_full or forward_and_recover_dues: NPL nets what the alternate partner owes it (EARNINGS_RECEIVABLE) against the funds forwarded, within the outstanding receivable, with the partner's agreement recorded. Post-collection outcome is via the own wallet or a refund to the sender.
- One settlement model: DISBURSEMENT drains one partner's holding into one or many DISBURSEMENT_LINES, in whole units. Under disburse_policy to_zero the balance after release is less than one unit; under hold_allowed the balance may rest at the partner within the project's exposure cap.

- A line's `paid_receiver_id` may be in another group: that is the offset, created by the same action, with Management approval and tighter aging.
- **Rejected payout.** When the receiving or intermediary bank rejects a transfer, `DISBURSEMENT_RETURN` records the returned amount, bank fee and reason; the line's outcome becomes returned, a reversing ledger transaction puts the money back into the client's balance and `BANK_FEE_EVENT` books the bounce fee. The money then stays in balance for later payouts, or funds a return leg that converts back and pays the sender's return receiver.
- **Fee decisions.** `FEE_DECISION` is a one-off, approved decision on a single subject: markup applied or waived on a return leg or balance conversion, a network-fee override on a refund, a bank-fee treatment override on one settlement. Standing rules stay in `FEE_STRUCTURE` and `FEE_OVERRIDE`.
- `CONFIRMATION` is full or short; a short confirmation opens a `SHORTFALL` topped up by a later disbursement. Rounding differences no longer land here. `REFERRAL_ACCRUAL` and `PARTNER_REBATE_ACCRUAL` accrue per deal under their rules; rebates are reconciled monthly against the partner's statement.

4 · Ledger & controls domain

Notes

- Client due balances are held as one account per project × partner × currency (CL.<PARTNER>. <CCY>.DUE); group entitlement is a view, not an account. holder_type never includes a settlement rail or narrative entity.
- **Ownership flips client → company at one event only: margin recognition at conversion.** NPL never advances money to a receiver, and withholding on a reroute is not a separate event because the margin is always captured by whichever partner converts.
- Recover dues on a reroute posts own-wallet cash against the earnings receivable at the partner: company money on both sides, no income.
- New transaction sources: BalanceConversion, DisbursementReturn, Referral, Rebate. New account purposes: REFERRAL_PAYABLE, REBATE_RECEIVABLE; ADVANCE is removed. New cost components: referral, rounding.
- APPROVAL, AUDIT_LOG and RECORD_LOCK are modelled once and apply to all master data. Approver ≠ initiator on every approval; Operations never sees derived economics. New approval actions: refund_to_third_party, cross_group_payout, balance_conversion, recover_dues.

5 · Invariants (enforced at write time)

1. Postings in one transaction sum to zero per currency.
2. Ledger tables are append-only; corrections are reversing transactions.
3. No custody in the normal path: CL.OWN.*.HELD may only be debited by a collection on a deal with an approved REROUTE; the custody aging view is the standing control.
4. Disburse to less than one unit: under disburse_policy = to_zero a DISBURSEMENT pays whole units and leaves only rounding dust; under hold_allowed the balance may rest at the partner within the project's exposure cap (Q5, Q8).
5. Priced once, per leg: a leg's pricing fields are write-once after priced_at; a re-quote is a new leg or a new deal group.
6. Partner fixed at quote, per leg: partner_pair_id / facilitating_entity_id change only through an approved REROUTE.
7. Entitlement conservation: $\Sigma \text{GROUP_ENTITLEMENT}$ over a project's groups = $\Sigma \text{CLIENT_BALANCE}$ over partners, per currency, including balance-conversion re-attributions.
8. Destination approval precedes collection: a quote-first deal cannot leave Inquiry unless at least one account in the entitled group has an approved SETTLEMENT_REGISTRATION with the chosen partner.
9. Approver $\neq$ initiator on every APPROVAL; Operations never receives derived-economics fields.
10. Every balance shown anywhere is a sum over postings or deals.
11. A collect-first deal requires an existing SENDER_RECEIVER_ALLOW entry for its sender and group (Q1).
12. A return leg pays the sender_return RECEIVER linked to the parent leg's sender, unless a Management-approved exception with a reason names another approved receiver.
13. The failed forward line is reversed before a return leg opens, so entitlement and balance are correct at the return conversion.
14. A CONVERSION belongs to exactly one owner: a deal leg or a balance conversion.
15. A recovery on a reroute cannot exceed the outstanding EARNINGS_RECEIVABLE at that partner; the alternate partner's expected conversion is computed on the net amount forwarded.
16. Ownership flips client $\rightarrow$ company at one event only: margin recognition at conversion. NPL never advances money to a receiver (Q13).
17. Offsets exist only across receiver groups and carry Management approval and tighter aging; within a group nothing is tracked beyond an informational tally.

6 · What changed from v3.0

12 new tables, 16 changed tables, 3 removals and one view replaced. Field-level detail is in Appendix B.

New tables

Table	Purpose
INTRODUCER	An external introducer who brought in a client and is owed referral commission (Understanding §4.2a). Paid either a share of NPL's earnings or a fixed % per transaction; several introducers may share one deal.

Table	Purpose
RECEIVER_GROUP	The unit entitlement is tracked on (design F). Every project has one default group created with it; receivers sit in the default group unless further groups are created and receivers are allocated to them. Payouts within a group are ordinary settlement; across groups they are offsets. With a single group no offset can arise. Each project also has a group of kind return that holds its sender_return receivers.
RECEIVING_ENTITY_ACCOUNT	One account of a receiving entity, with its own currency and rail (Q4: EUR via SWIFT and EUR via SEPA are two accounts). Registrations and payout lines point at the account. A sender_return receiver's USDT wallet is an account like any other.
REFERRAL_RULE	How an introducer is paid on a project (and optionally only on one partner × pair): a share of NPL's earnings in a ratio, or a fixed % applied inside the sender-facing markup (§4.2a).
PARTNER_REBATE_RULE	A partner rebating part of its fee to NPL (§4.2b: Jeton charges 1.25% and gives back 0.4%). Reconciled monthly through PARTNER_REBATE_ACCRUAL.
DEAL_GROUP	The client-facing instruction the legs fulfil: the pair as the sender sees it (INR → EUR), the sender rate, fee and reference. Every deal belongs to a group; a plain single-partner deal is a group of one leg. Margin across a two-leg route is the group quote minus the partner conversions.
BALANCE_CONVERSION	NPL converting a client's held balance at a partner (5K SGD → USDT to fund a 15K USDT payout). Not a deal leg: no sender, receiver or collection; Finance requests, Management approves. Uses the same rate machinery; markup is a FEE_DECISION. Names the receiver group it serves (chosen split).
ENTITLEMENT_REATTRIBUTION	Which group's entitlement moved currency in a balance conversion, so entitlement per currency still equals balance per currency (invariant 7). The excess over the served group's source-currency entitlement is another group's flow and is booked as an approved offset.
DISBURSEMENT_RETURN	A payout line the receiving or intermediary bank rejected; the money came back to the partner minus bank fees. Re-enters the balance by a reversing ledger transaction; the fee is booked through BANK_FEE_EVENT. May fund a return leg (funding_source = returned_disbursement) or simply stay in balance for later payouts.
FEE_DECISION	A one-off, approved fee decision on a single subject: markup applied or waived on a return leg or balance conversion, a network-fee override on a refund (Q7), a bank-fee treatment override on one settlement. Standing rules stay in FEE_STRUCTURE and FEE_OVERRIDE.
REFERRAL_ACCRUAL	Referral commission accrued to an introducer on one deal under a REFERRAL_RULE; paid out of NPL's earnings.
PARTNER_REBATE_ACCRUAL	Rebate expected from a partner on one deal, reconciled monthly against the partner's statement and received (usually in crypto). Feeds the Partner Commissions report.

Changed tables

Table	Change
PROJECT	+ network_fee_policy (Q7) + disburse_policy (Q5); exposure_ceiling now the cap on held balance receiver_fee_timing default moves to at_conversion (Q6)

Table	Change
RECEIVER	+ group_id (D1) + kind, sender_id, partner_config_id (D3) sender_return receivers never appear on forward deals or allow lists
OWN_WALLET	+ provider, account_label (Q10)
CURRENCY_PAIR	+ amount_rounding, cash_rounding_unit (Q8, Sud)
SENDER_RECEIVER_ALLOW	receiver_id → receiver_group_id (D1) rule: must exist before a collect-first deal (Q1)
RECEIVING_ENTITY	– bank_details_ref, – currency (moved to RECEIVING_ENTITY_ACCOUNT, Q4)
SETTLEMENT_REGISTRATION	receiving_entity_id → receiving_entity_account_id (Q4)
DEAL	+ deal_group_id, leg_no, leg_type, funding_source, funding_ref_id (D2) + receiver_group_id; receiver_id optional (D1) deal_ref and entry_mode move to DEAL_GROUP quote clocks nullable on conversion-priced legs
CONVERSION	owner is a deal leg or a balance conversion (D4) + rounding_residual (Q8) + fee_receiver_part (Q6)
REROUTE	forwarding_policy: withhold_earnings removed (Q2) + recovered_amount, partner_agreement_ref (recover dues) + outcome for post-collection reroutes (Q3)
DISBURSEMENT	invariant 4 restated: to less than one unit, or held within the exposure cap (Q5, Q8)
DISBURSEMENT_LINE	+ outcome incl. returned (D6) registration now points at an account (Q4)
SHORTFALL	no longer receives rounding differences (Q8)
LEDGER_TRANSACTION	+ source types BalanceConversion, DisbursementReturn, Referral, Rebate
LEDGER_POSTING	+ cost components referral, rounding ownership flips at one event only (Q2, Q13)
LEDGER_ACCOUNT	– purpose ADVANCE (Q13) + purposes PAYABLE, REFERRAL_PAYABLE, REBATE_RECEIVABLE + holder_type Introducer, Project (PAYABLE) WALLET also used with owner company for NPL's own wallet money (ledger design v1.0)

Removed

Item	Reason
RECEIVER_ENTITLEMENT (view)	Replaced by GROUP_ENTITLEMENT: entitlement is tracked per receiver group.
REROUTE.forwarding_policy = withhold_earnings	Removed: NPL's margin is always captured at conversion.
LEDGER_ACCOUNT purpose ADVANCE, and the advance approval	Removed: NPL never advances money to a receiver.

Views

View	Change
CLIENT_BALANCE	includes balance conversions and returns; carries dust
GROUP_ENTITLEMENT	replaces RECEIVER_ENTITLEMENT (D1)
OFFSET	per group, not per receiver; no tiers (D1)
CUSTODY	includes recovered dues

7 · Open questions

2 of the questions carried from the Understanding document remain open; the rest were answered by NPL in the review of the Understanding document or settled in the ERD review of 6–7 October 2026 and are reflected in this draft.

Still open

#	Question	Blocks	Status
9	FRS figure reconciliation (1,208 vs 1,225; Mode 1 example)	Golden cases	Sud: Needs clarification. To be discussed.
11	Reference workbooks per project archetype	Calculation acceptance	Sud: Will be added soon.

Answered and reflected in v4.0

#	Question	Answer	In the model
1	Collect-first: must SENDER_RECEIVER_ALLOW pre-exist for attribution; handling of inbound matching no sender	Sud: Yes, sender-receiver mapping should be done in configuration. When the transaction is created, the sender is picked from a list.	v4.0: write-time rule that a collect-first deal needs an existing allow entry; the allow list now targets receiver groups.
2	Withholding-eligibility criteria on rerouted deals	Review (7 Oct): NPL's cut is always captured at conversion, so it is never at risk from a return or a forward. If forwarding happens before conversion, whichever partner converts captures it.	v4.0: the eligibility question dissolves. forwarding_policy is forward_full or forward_and_recover_dues; withhold_earnings is removed. Ownership flips at one event only.
3	Post-collection reroute (funds stuck at a partner): transfer, offset, or refund-and-requote	Sud: The partner can send the funds to NPL (and NPL decides where to send) or the partner can refund the sender.	v4.0: REROUTE.outcome via_own_wallet / refund_sender; a refund is a return leg paying the sender_return receiver.

#	Question	Answer	In the model
4	Receiving-entity constraints (currency, rail) narrowing settlement choice	Sud: Receiving entities have specific bank accounts with currency and rail restrictions, for example EUR via SWIFT and EUR via SEPA.	v4.0: RECEIVING_ENTITY_ACCOUNT; registrations and payout lines point at the account.
5	Any deliberate partial-disbursement / reserve policy	Sud: Yes, possible. In some projects the client's funds sit until they are ready to receive them, within an exposure threshold NPL sets by appetite.	v4.0: PROJECT.disburse_policy to _zero / hold_allowed with exposure_ceiling as the cap; invariant 4 restated.
6	Receiver-side fee timing per project	Sud: Receiver-side fee should be applied at the time of conversion, otherwise the balance will be incorrect.	v4.0: default at _conversion; CONVERSION.fee_receiver_part; DISBURSEMENT_LINE.receiver_fee_applied becomes exceptional.
7	Network-fee policy on refunds	Sud: Set at project level with per-transaction override (one 100 USD fee was repaid in the next settlement).	v4.0: PROJECT.network_fee_policy plus FEE_DECISION for the override.
8	Whole-unit conversion rounding rule	Review (7 Oct): fractions are dropped to whole units at conversion, per transaction (30.152 → 30). The dropped fraction stays client money, carried in the balance and paid out when it reaches a whole unit. Cash collections round to the nearest 100 / 500 / 1000 by currency.	v4.0: CONVERSION.rounding_residual; CURRENCY_PAIR.amount_rounding and cash_rounding_unit; disburse to less than one unit; SHORTFALL no longer takes rounding.
10	AQN-LT-Sub “settled by offset against transactions outside the system”	Sud: This is NPL's wallet (pass-through at collection or on the settlement side). AQN is Aquanow, a crypto exchange; LT Sub is the account name.	v4.0: OWN_WALLET.provider and account_label; pass-through is a two-leg route or a reroute via the own wallet.
12	FRS P1 / P3 vs rerouting and multi-partner use — amend FRS or restrict scope	Sud: Check the added explanations (partner types and channels; pass-through and double conversions in §7.1.1). Ask again if still unclear.	v4.0: two-leg routes are deal groups with two legs and a partner_transit receiver for the hop; FRS principles generalised, scope kept.

#	Question	Answer	In the model
13	Advance funding source	Review (7 Oct): never. NPL does not pay a receiver before the client's money has arrived.	v4.0: LEDGER_ACCOUNT purpose ADVANCE removed; the advance approval and the "advance" ownership-flip event removed.
14	Review point: entitlement split on a balance conversion	Review (7 Oct): chosen group, not proportional. The conversion is triggered by a specific payout.	v4.0: BALANCE_CONVERSION.receiver_group_id; excess over the group's source-currency entitlement is booked as an approved offset.
15	Review point: refunds to a third party at the sender's request	Review (7 Oct): allowed by exception only. Default is the sender's own return receiver; any other destination needs Management approval with a reason and a pre-registered approved receiver.	v4.0: invariant on return legs; APPROVAL action refund_to_third_party.
16	Review point: recover dues on a reroute	Review (7 Oct): yes, NPL nets what a partner owes it against funds about to be forwarded to that partner.	v4.0: forwarding_policy forward_and_recover_dues, recovered_amount, partner_agreement_ref; postings own wallet cash ← earnings receivable, no income.
17	Review point: legs of a route share one client-facing quote	Review (7 Oct): yes, a deal group. The sender saw one price and one reference; margin across legs is the group quote minus the partner conversions.	v4.0: DEAL_GROUP; every deal belongs to a group, a single-partner deal being a group of one leg.

Appendix A · FRS reconciliation

A.1 Vocabulary

FRS term	Model entity
Client	PROJECT
Receiving Entity	RECEIVING_ENTITY (member of a RECEIVER, which belongs to a RECEIVER_GROUP) with its RECEIVING_ENTITY_ACCOUNTs
Sender	SENDER with SENDER_RECEIVER_ALLOW
Partner (PT, Jeton)	PARTNER + PARTNER_CONFIG
Wallet registry	WALLET under FACILITATING_ENTITY
Rate-day versions	RATE_COMPARISON + PARTNER_RATE_VERSION
Settlement	DISBURSEMENT + DISBURSEMENT_LINE + CONFIRMATION (+ DISBURSEMENT_RETURN when rejected)
Shortfall ledger	SHORTFALL + CL.<RECEIVER>.<CCY>.SHORTFALL
PT Comms / H Comms	C0.<PARTNER>.<CCY>.POOL (earnings receivable)
Inter-entity redirection	Offset across receiver groups: DISBURSEMENT_LINE.paid_receiver_id in another group than the entitled one
Double conversion / pass-through	DEAL_GROUP with two legs (funding_source = prior_leg), hop via a partner_transit RECEIVER

A.2 Deviations from the FRS (classified)

FRS statement	Model	Classification
P1 — no custody	No custody in the normal path; OWN_WALLET + REROUTE for contingency	Decision required (Q12)
P3 — partner books never mix	Accounts per partner never net; alternate partner on reroute; post-collection transfer only if Q3 confirms	Decision required (Q12)
Strict Client → Entity → Sender, one parent each	RECEIVER_GROUP → RECEIVER → RECEIVING_ENTITY → account; SENDER with a group allow-list; FRS project = one default group	Generalisation per NPL instruction
Beyond grace → re-quote always	late_collection_policy per project; FRS project keeps requote	Per NPL instruction (Sept 2026)
Sender always priced on Kraken	FEE_STRUCTURE.rate_basis; FRS project = market	Generalisation
One settlement per deal, full amount	Accumulate → disburse to less than one unit (whole units; rounding dust carried), or hold within the exposure cap where the project allows	Generalisation
Partner fixed at quote	Same, except under REROUTE	Consistent
Bank fee → shortfall → top-up	carry_forward is the FRS behaviour; waived / absorbed added	Per NPL instruction
Registry-only destinations	SETTLEMENT_REGISTRATION.approval_status — FRS project has all approved	Generalisation
AQN-LT-Sub offset-settled collections	OWN_WALLET at Aquanow; pass-through is a two-leg deal group or a reroute via the own wallet	Resolved (v4.0)

Draft v4.0 — for review with NPL. Dependent documents (ledger design, calculation spec, state machines, roles matrix, configuration schema, cutover) to be rebuilt against this version.

Appendix B · Data dictionary (field annotations)

Every table also carries `created_at` and `created_by`. Types: `int`, `string`, `bool`, `decimal`, `datetime`, `date`.

B.1 Configuration

PROJECT

Field	Type	Key	Notes
id	int	PK	
name	string		
client_label	string		
quote_freshness_min	int		
quote_validity_min	int		
grace_min	int		
grace_drift_pct	decimal		
late_collection_policy	string		requote / honor_with_approval
bank_fee_treatment_default	string		waived / carry_forward / absorbed
receiver_fee_timing	string		at_conversion (default, Q6) / post_confirmation
network_fee_policy	string		absorbed / charged_back (project default; per-transaction override via FEE_DECISION)
disburse_policy	string		to_zero / hold_allowed (Q5)
shortfall_tolerance	decimal		
exposure_ceiling	decimal		cap on balance resting at a partner
status	string		

PARTNER

Field	Type	Key	Notes
id	int	PK	
name	string		Ali, Jeton, PT
disclosure_style	string		markup_included / market_plus
rate_expiry_time	string		
holiday_calendar	string		
status	string		

RATE_SOURCE

Field	Type	Key	Notes
id	int	PK	
code	string		KRAKEN / CMC / OANDA / XE
staleness_threshold_s	int		

INTRODUCER

Field	Type	Key	Notes
id	int	PK	
name	string		
status	string		

SENDER

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
name	string		Ayush, Naveen
risk_tier	string		
status	string		

RECEIVER_GROUP

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
name	string		
is_default	bool		exactly one per project
kind	string		client / return
status	string		

RECEIVER

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
group_id	int	FK	receiver group
kind	string		counterparty / sender_return / partner_transit
sender_id	int	FK	when kind = sender_return
partner_config_id	int	FK	when kind = partner_transit
name	string		the group: Sud, Raj
status	string		

OWN_WALLET

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
provider	string		e.g. Aquanow (Q10)
account_label	string		e.g. LT Sub
network	string		
address	string		
signer_arrangement	string		hardware / multisig
custody_max_hours	int		

THRESHOLD

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
kind	string		max_amount / settlement_limit / daily_slots / rate_variation_rel / custom

Field	Type	Key	Notes
currency	string		
value	decimal		
condition_expr	string		custom conditions
escalation	string		finance / management

CURRENCY_PAIR

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
from_currency	string		
to_currency	string		
quote_direction	string		out_per_in / in_per_out
rate_precision	int		
amount_rounding	string		truncate_whole_unit (Q8)
cash_rounding_unit	int		100 / 500 / 1000 by currency
status	string		

PARTNER_CONFIG

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
partner_id	int	FK	
applicable_for_reroute	bool		
status	string		

COLLECTION_SENDING_ENTITY

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
label	string		sender's own account or a client's account
account_ref	string		

SENDER_RECEIVER_ALLOW

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
receiver_group_id	int	FK	sendable_to (was receiver_id)

RECEIVING_ENTITY

Field	Type	Key	Notes
id	int	PK	
receiver_id	int	FK	
name	string		NewXP Entity, ReferScout Entity

Field	Type	Key	Notes
status	string		

PAIR_RATE_SOURCE

Field	Type	Key	Notes
id	int	PK	
currency_pair_id	int	FK	
rate_source_id	int	FK	
priority	int		

PARTNER_PAIR

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
currency_pair_id	int	FK	
partner_pricing	string		market_plus_pct / disclosed_rate
partner_markup_pct	decimal		known only for market_plus
status	string		

FACILITATING_ENTITY

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
name	string		PT-tour, PT Global Inc
bank_details_ref	string		optional
status	string		

COLLECTION_RECEIVING_ENTITY

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
name	string		
bank_details_ref	string		
currency	string		
status	string		

SETTLEMENT_SENDING_ENTITY

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
narrative_name	string		what the receiver sees
status	string		

SETTLEMENT_RAIL

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
name	string		bank or EMI executing the transfer
bank_details_ref	string		
expected_bank_fee	decimal		
currency	string		

FEE_OVERRIDE

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	
partner_pair_id	int	FK	
pct	decimal		
effective_from	datetime		
approved_by	int	FK	

RECEIVING_ENTITY_ACCOUNT

Field	Type	Key	Notes
id	int	PK	
receiving_entity_id	int	FK	
currency	string		
rail	string		SWIFT / SEPA / TRC-20 / ERC-20 / ...
details_ref	string		bank or wallet details
status	string		

SETTLEMENT_REGISTRATION

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
receiving_entity_account_id	int	FK	was receiving_entity_id
approval_status	string		pending_partner / approved / suspended
bank_details_ref	string		as paid by this partner
approved_at	datetime		

FEE_STRUCTURE

Field	Type	Key	Notes
id	int	PK	
partner_pair_id	int	FK	
rate_basis	string		market / partner
mode	string		fixed / variable
pct	decimal		
sender_share_pct	decimal		

Field	Type	Key	Notes
receiver_share_pct	decimal		
floor_pct	decimal		
cap_pct	decimal		
min_margin_pct	decimal		
variance_threshold_rel	decimal		
quoting_rate_policy	string		worst_applicable / designated / best_applicable
effective_from	datetime		

WALLET

Field	Type	Key	Notes
id	int	PK	
facilitating_entity_id	int	FK	
network	string		TRC-20 / ERC-20
address	string		
currency	string		
status	string		

REFERRAL_RULE

Field	Type	Key	Notes
id	int	PK	
introducer_id	int	FK	
project_id	int	FK	
partner_pair_id	int	FK	optional
basis	string		share_of_earnings / fixed_pct
pct	decimal		
effective_from	datetime		

PARTNER_REBATE_RULE

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
pct_of_partner_fee	decimal		
effective_from	datetime		

B.2 Rates

MARKET_RATE

Field	Type	Key	Notes
id	int	PK	
rate_source_id	int	FK	
currency_pair_id	int	FK	
rate	decimal		
captured_at	datetime		

RATE_COMPARISON

Field	Type	Key	Notes
id	int	PK	
partner_pair_id	int	FK	
rate_date	date		
avg_spread_2w	decimal		
avg_spread_4w	decimal		
status	string		open / closed
created_at	datetime		created lazily by the first deal of the day

PARTNER_RATE_VERSION

Field	Type	Key	Notes
id	int	PK	
rate_comparison_id	int	FK	
version_no	int		
rate	decimal		
market_rate_id	int	FK	
spread_pct	decimal		
status	string		active / usable / breached / approved / rejected
confirmed_by	int	FK	
approval_reason	string		
expires_at	datetime		

B.3 Deal, collection, conversion, reroute, balance conversion

DEAL_GROUP

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
sender_id	int	FK	
receiver_group_id	int	FK	entitled group
client_ref	string		D-YYMMDD-seq, what the sender sees
from_currency	string		
to_currency	string		
sender_rate	decimal		
fee_structure_id	int	FK	
amount_in	decimal		
target_out	decimal		
entry_mode	string		quote_first / collect_first / estimate_first
state	string		
quoted_at	datetime		

DEAL

Field	Type	Key	Notes
id	int	PK	
deal_group_id	int	FK	every deal belongs to a group
leg_no	int		
leg_type	string		forward / return
funding_source	string		collection / returned_disbursement / prior_leg
funding_ref_id	int		DISBURSEMENT_RETURN id or prior DEAL id
project_id	int	FK	
sender_id	int	FK	
receiver_group_id	int	FK	attribution (D1)
receiver_id	int	FK	optional until settlement
partner_pair_id	int	FK	
facilitating_entity_id	int	FK	
amount_mode	string		target_out / fixed_in
collection_method	string		crypto / bank / cash
cash_instructions	string		
fee_structure_id	int	FK	
market_rate_id	int	FK	
partner_rate_version_id	int	FK	
sender_rate	decimal		
amount_in	decimal		
gross_out	decimal		
client_net	decimal		
fee_sender_part	decimal		
fee_receiver_part	decimal		
priced_at	datetime		quote or conversion
sent_at	datetime		null on conversion-priced legs
valid_until	datetime		
grace_until	datetime		
late_decision	string		none / honored / requoted
state	string		
quote_hash	string		

COLLECTION

Field	Type	Key	Notes
id	int	PK	
deal_id	int	FK	
endpoint_type	string		wallet / collection_receiving_entity / cash
endpoint_id	int		
collection_sending_entity_id	int	FK	
amount	decimal		
evidence_ref	string		hash / slip / receipt

Field	Type	Key	Notes
state	string		
verified_by	int	FK	
verified_at	datetime		

CONVERSION

Field	Type	Key	Notes
id	int	PK	
deal_id	int	FK	null when owned by a balance conversion
balance_conversion_id	int	FK	null when owned by a deal leg
partner_rate_version_id	int	FK	
calc_mode	string		npl_calculates / partner_calculates
amount_in	decimal		
expected_out	decimal		
actual_out	decimal		booked figure, whole units
rounding_residual	decimal		fraction dropped; stays client money (Q8)
fee_receiver_part	decimal		taken at conversion (Q6)
variance	decimal		
residual_in	decimal		margin pool
basis_variance	decimal		
state	string		
approved_by	int	FK	

REROUTE

Field	Type	Key	Notes
id	int	PK	
deal_id	int	FK	
trigger	string		partner_unavailable / limit_reached / holiday / compliance_hold / other
phase	string		pre_collection / post_collection
outcome	string		via_own_wallet / refund_sender (Q3, post-collection)
own_wallet_id	int	FK	
alternate_partner_config_id	int	FK	
forwarding_policy	string		forward_full / forward_and_recover_dues
recovered_amount	decimal		≤ outstanding receivable at the alternate partner
partner_agreement_ref	string		the partner agreed the net amount
reason	string		
requested_by	int	FK	
approved_by	int	FK	

SENDER_CREDIT

Field	Type	Key	Notes
id	int	PK	
sender_id	int	FK	

Field	Type	Key	Notes
deal_id	int	FK	
amount	decimal		
state	string		open / applied

EARNINGS_RECEIVABLE

Field	Type	Key	Notes
id	int	PK	
partner_config_id	int	FK	
deal_id	int	FK	
amount	decimal		
state	string		outstanding / recovered
recovered_by_deal_id	int	FK	
recovered_at	datetime		

BALANCE_CONVERSION

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
partner_config_id	int	FK	
receiver_group_id	int	FK	group served; chosen split
from_currency	string		
to_currency	string		
partner_pair_id	int	FK	
partner_rate_version_id	int	FK	
market_rate_id	int	FK	
amount_in	decimal		
expected_out	decimal		
actual_out	decimal		
variance	decimal		
rounding_residual	decimal		
state	string		
requested_by	int	FK	Finance
approved_by	int	FK	Management
converted_at	datetime		

ENTITLEMENT_REATTRIBUTION

Field	Type	Key	Notes
id	int	PK	
balance_conversion_id	int	FK	
receiver_group_id	int	FK	
amount_from	decimal		
amount_to	decimal		

B.4 Disbursement, return, confirmation, fees, referrals, rebates

DISBURSEMENT

Field	Type	Key	Notes
id	int	PK	
project_id	int	FK	
partner_config_id	int	FK	
settlement_sending_entity_id	int	FK	
settlement_rail_id	int	FK	
currency	string		
total	decimal		whole units
bank_fee_treatment	string		
state	string		
prepared_by	int	FK	
approved_by	int	FK	
escalated_to	int	FK	
released_at	datetime		

DISBURSEMENT_LINE

Field	Type	Key	Notes
id	int	PK	
disbursement_id	int	FK	
settlement_registration_id	int	FK	
paid_receiver_id	int	FK	
amount	decimal		
receiver_fee_applied	decimal		exceptional; fee is normally taken at conversion
reference	string		entity invoice format
outcome	string		sent / confirmed / returned

DISBURSEMENT_RETURN

Field	Type	Key	Notes
id	int	PK	
disbursement_line_id	int	FK	
returned_amount	decimal		
bank_fee	decimal		
reason	string		
returned_at	datetime		

CONFIRMATION

Field	Type	Key	Notes
id	int	PK	
disbursement_line_id	int	FK	
amount_received	decimal		
slip_ref	string		

Field	Type	Key	Notes
result	string		full / short
confirmed_by	int	FK	
confirmed_at	datetime		

SHORTFALL

Field	Type	Key	Notes
id	int	PK	
receiver_id	int	FK	
confirmation_id	int	FK	
amount	decimal		
state	string		open / topped_up
topped_up_disbursement_id	int	FK	

BANK_FEE_EVENT

Field	Type	Key	Notes
id	int	PK	
disbursement_id	int	FK	
expected_amount	decimal		
actual_amount	decimal		
treatment	string		waived / carry_forward / absorbed
state	string		open / applied
applied_disbursement_id	int	FK	

FEE_DECISION

Field	Type	Key	Notes
id	int	PK	
subject_type	string		Deal / Disbursement / BalanceConversion
subject_id	int		
fee_kind	string		markup / network / bank
decision	string		apply / waive / override
pct	decimal		
approved_by	int	FK	
reason	string		

REFERRAL_ACCRUAL

Field	Type	Key	Notes
id	int	PK	
referral_rule_id	int	FK	
deal_id	int	FK	
amount	decimal		
state	string		accrued / paid
paid_at	datetime		

PARTNER_REBATE_ACCRUAL

Field	Type	Key	Notes
id	int	PK	
partner_rebate_rule_id	int	FK	
deal_id	int	FK	
expected_amount	decimal		
received_amount	decimal		
period	string		YYYY-MM
state	string		expected / reconciled / received

B.5 Ledger & controls

LEDGER_TRANSACTION

Field	Type	Key	Notes
id	int	PK	
source_type	string		Collection / Conversion / BalanceConversion / Disbursement / DisbursementReturn / Confirmation / BankFeeEvent / Reroute / Referral / Rebate / Adjustment
source_id	int		
description	string		
posted_at	datetime		
posted_by	int	FK	
reversal_of_id	int	FK	

LEDGER_POSTING

Field	Type	Key	Notes
id	int	PK	
ledger_transaction_id	int	FK	
ledger_account_id	int	FK	
side	string		debit / credit
amount	decimal		
currency	string		
ownership_tag	string		client / company
cost_component	string		principal / earnings / partner_cost / bank_fee / network_fee / variance / referral / rounding

LEDGER_ACCOUNT

Field	Type	Key	Notes
id	int	PK	
code	string		OWNER.HOLDER.CCY.PURPOSE
ownership	string		client / company
holder_type	string		PartnerConfig / OwnWallet / Receiver / Sender / Introducer / Project / None
holder_id	int		
currency	string		

Field	Type	Key	Notes
purpose	string		COLLECTED / HELD / DUE / INTRANSIT / PAYABLE / SHORTFALL / CREDIT / POOL / WALLET / EARN_GROSS / EXP_PARTNER / EXP_BANKFEE / EXP_NETWORK / VARIANCE / REFERRAL_PAYABLE / REBATE_RECEIVABLE
kind	string		asset / liability / income / expense

USER

Field	Type	Key	Notes
id	int	PK	
email	string		Google Workspace
role	string		ops / finance / management
status	string		

APPROVAL

Field	Type	Key	Notes
id	int	PK	
subject_type	string		
subject_id	int		
action	string		
requested_by	int	FK	
approved_by	int	FK	never equal to requested_by
escalated_to	int	FK	
state	string		
reason	string		

AUDIT_LOG

Field	Type	Key	Notes
id	int	PK	
record_type	string		
record_id	int		
field	string		
old_value	string		
new_value	string		
actor_id	int	FK	
at	datetime		

RECORD_LOCK

Field	Type	Key	Notes
id	int	PK	
record_type	string		
record_id	int		
locked_by	int	FK	
locked_at	datetime		